# Summer Project

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April 10, 2026

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# 1 Background Information

## 1.1 Optical Physics

The optical beam is modelled as a Gaussian beam propagating along the optical axis ($z$-axis). Working in the paraxial approximation, the complex electric field is

$$E(x,y,z) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{x^2+y^2}{w(z)^2}\right) \exp\left(-ikz - i\frac{k(x^2+y^2)}{2R(z)} + i\zeta(z)\right), \tag{1}$$

where $k = 2\pi/\lambda$ is the wavenumber and the beam parameters are

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}, \tag{2}$$

$$R(z) = z\left(1+\left(\frac{z_R}{z}\right)^2\right), \tag{3}$$

$$\zeta(z) = \arctan\left(\frac{z}{z_R}\right), \qquad z_R = \frac{\pi w_0^2}{\lambda}. \tag{4}$$

Here $w_0$ is the beam waist, $w(z)$ is the beam radius at position $z$, $R(z)$ is the radius of curvature of the wavefronts, $\zeta(z)$ is the Gouy phase, and $z_R$ is the Rayleigh range.

At the SLM plane $z = z_0$ the amplitude prefactor $w_0/w(z_0)$ is constant, and $\exp(-ikz_0)$ and $\exp(i\zeta(z_0))$ are *global* phases, so they may be dropped. The curvature term

$$\exp\left(-i\frac{k(x^2+y^2)}{2R(z_0)}\right)$$

is spatially varying and should be retained unless the beam is well collimated. The incident field at the SLM plane is therefore

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right)\exp\left(-i\frac{k(x^2+y^2)}{2R}\right), \tag{5}$$

where $w \equiv w(z_0)$, $R \equiv R(z_0)$, and $A$ absorbs all constant prefactors. In the collimated limit ($R \to \infty$) this reduces to

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right). \tag{6}$$

The modulated field then propagates through thermo-optical material modelled by a linear transmission matrix $T$, and the detector measures only intensity:

$$E_{\text{out}} = T\,E_{\text{SLM}}, \qquad I(x,y) = |E_{\text{out}}(x,y)|^2. \tag{7}$$

Optical detectors measure intensity, not complex amplitude, so $I(x,y)$ is the only accessible quantity.

In summary, the incident field is the Gaussian beam evaluated at the SLM plane; spatially varying curvature phase is retained when needed; the downstream optical medium is captured by a complex transmission matrix; and the output camera measures intensity.

## 1.2 Thermal Physics

The optical medium is treated as a thermally perturbed phase/scattering element whose temperature field evolves on timescales much slower than the optical field. Over these slower thermal timescales, the temperature distribution $T(\mathbf{r}, t)$ is modelled by the heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q(\mathbf{r}, t), \tag{8}$$

where $\rho$ is the density, $c_p$ is the specific heat capacity, $\kappa$ is the thermal conductivity, and $Q(\mathbf{r}, t)$ is any applied or absorbed heat source.

Temperature affects the optical response through the thermo-optic effect. To first order, the refractive index is taken to vary linearly with temperature:

$$n(\mathbf{r}, t) = n_0 + \frac{dn}{dT}(T(\mathbf{r}, t) - T_0), \tag{9}$$

where $n_0$ is the reference refractive index at temperature $T_0$, and $dn/dT$ is the thermo-optic coefficient of the material. These refractive-index variations change the optical path length through the medium and therefore introduce a time-dependent phase perturbation

$$\Delta\phi(x, y, t) = k \int \Delta n(x, y, z, t)\, dz, \tag{10}$$

with $k = 2\pi/\lambda$ and $\Delta n = n - n_0$.

At the system level, these thermally induced phase changes are represented as a slow drift in the effective optical operator of the medium. In the transmission-matrix description used here, this is written as

$$T_{\text{opt}}(t) = T_{\text{opt},0} + \Delta T_{\text{opt}}(t), \tag{11}$$

where $T_{\text{opt},0}$ is the nominal transmission matrix and $\Delta T_{\text{opt}}(t)$ captures the time-varying perturbation caused by thermal diffusion. The key thermal-physics problem is therefore to model how heat input produces a temperature field, how that temperature field modifies refractive index, and how the resulting refractive-index changes alter the optical response seen by the detector.

## 1.3 SLM Behaviour

The SLM acts only in the transverse $(x, y)$ plane and is modeled as a phase-only modulator. The applied phase pattern $\phi(x, y)$ is the control input and sets the relative phases across the aperture, thereby altering interference at the output. The modulated field is

$$E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y) e^{i\phi(x,y)}.$$

## 1.4 Physics-Informed Neural Networks

Physics-informed neural networks (PINNs) are a class of neural-network models in which the unknown solution of a physical system is represented by a trainable function and constrained not only by data, but also by the governing differential equations of the system. Rather than learning an arbitrary input–output map, a PINN seeks a function that is consistent with both observations and the known underlying physics.

In the present context, the relevant unknown field is the temperature distribution $T(\mathbf{r}, t)$ inside the thermo-optical material. A PINN represents this field by a neural network $T_\theta(\mathbf{r}, t)$, where $\theta$

denotes the trainable parameters. The network is then trained so that its output satisfies the heat equation

$$\rho c_p \frac{\partial T_\theta}{\partial t} - \nabla \cdot (\kappa \nabla T_\theta) - Q(\mathbf{r}, t) = 0 \tag{12}$$

at a set of collocation points in space and time, while also matching any available initial conditions, boundary conditions, and measurement data.

The training objective therefore consists of several terms. A typical PINN loss takes the form

$$\mathcal{L} = \lambda_{\text{data}} \mathcal{L}_{\text{data}} + \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}}, \tag{13}$$

where $\mathcal{L}_{\text{data}}$ penalizes disagreement with observed data, $\mathcal{L}_{\text{PDE}}$ penalizes violations of the governing heat equation, $\mathcal{L}_{\text{BC}}$ enforces boundary conditions, and $\mathcal{L}_{\text{IC}}$ enforces the initial temperature field. The PDE residual is evaluated using automatic differentiation, which allows spatial and temporal derivatives of the neural-network output to be computed directly during training.

Once a thermally consistent approximation $T_\theta(\mathbf{r}, t)$ has been learned, it can be coupled to the optical model through the thermo-optic relation

$$n_\theta(\mathbf{r}, t) = n_0 + \frac{dn}{dT}\big(T_\theta(\mathbf{r}, t) - T_0\big), \tag{14}$$

which gives a corresponding refractive-index field. This refractive-index variation induces a time-dependent optical phase perturbation,

$$\Delta\phi_\theta(x, y, t) = k \int \big(n_\theta(x, y, z, t) - n_0\big)\, dz, \tag{15}$$

and hence modifies the effective optical response of the medium. In this way, the PINN serves as a physics-constrained surrogate model for the thermal dynamics that drive drift in the optical system.

The main advantage of this approach is that it embeds known thermal physics directly into the learning process, reducing the need for large volumes of labelled data and producing predictions that remain consistent with the governing equations. For the present project, this provides a principled framework for learning the slow time evolution of the thermo-optical material and relating that evolution to the changing interference pattern observed at the detector.

## 2 Experimental Setup

A setup of our system is shown below in Figure 1.

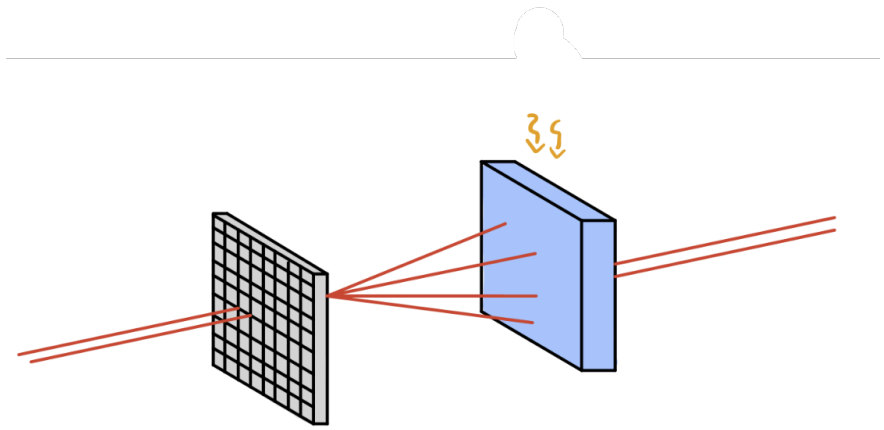

Figure 1: Schematic of the experimental setup. The laser beam is modulated by the SLM, passes through the some material, and is detected by the camera.

The optical process is summarized as follows:

i. Define the incident field $E_{\text{in}}(x, y)$ at the SLM plane.

ii. Apply SLM phase modulation to obtain $E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y)e^{i\phi(x,y)}$.

iii. Propagate through the thermo-optical material using the transmission matrix $T_{\text{opt}}(t)$ to give $E_{\text{out}} = T_{\text{opt}}(t)E_{\text{SLM}}$.

iv. Compute the output intensity $I(x, y) = |E_{\text{out}}(x, y)|^2$.

v. Define the reinforcement-learning reward as the intensity at a selected target output location.